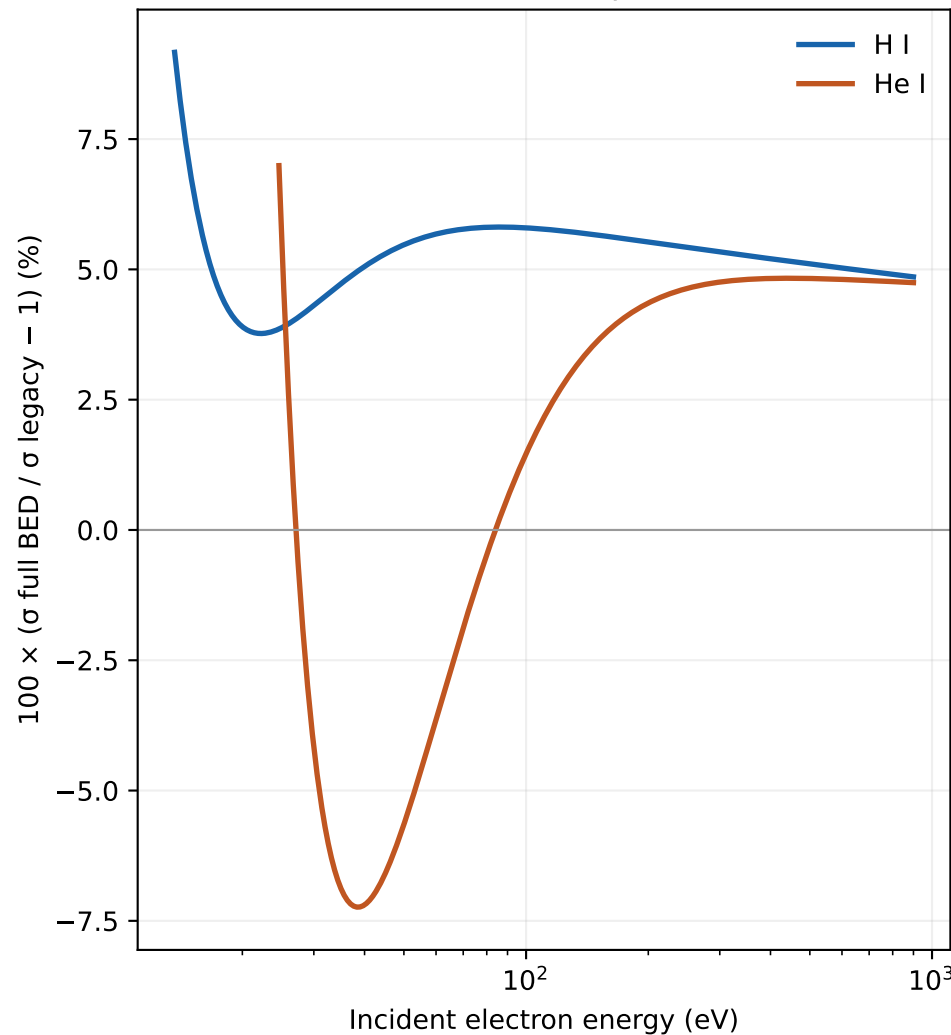
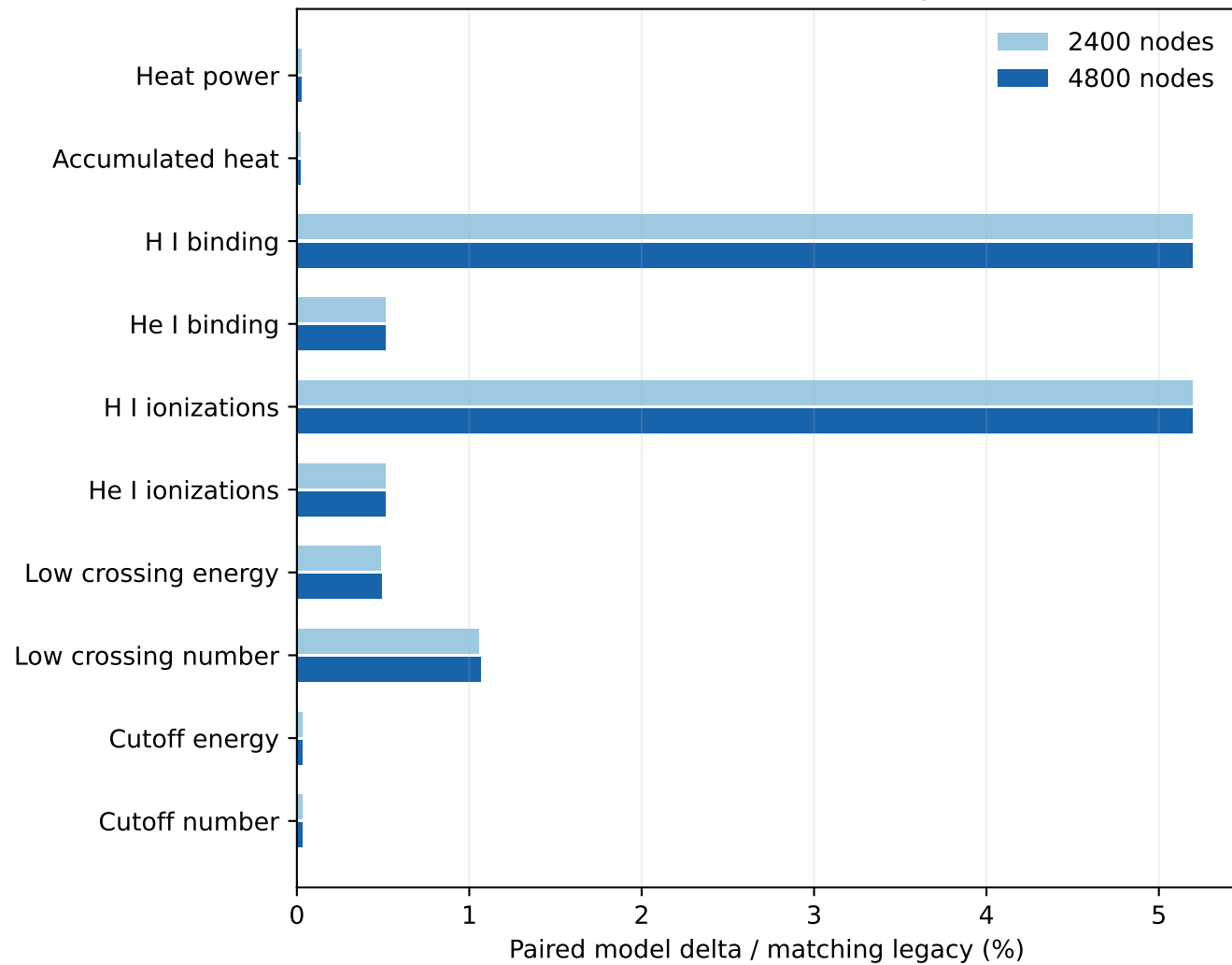


(a) Ionization total representation



(b) Finite-time cascade comparison



Fixed bath · selected 0.1-900 eV source · $T = 10^{10}$ s
Two-grid changes are empirical; no physical error certificate. Inherited global gates unchanged.